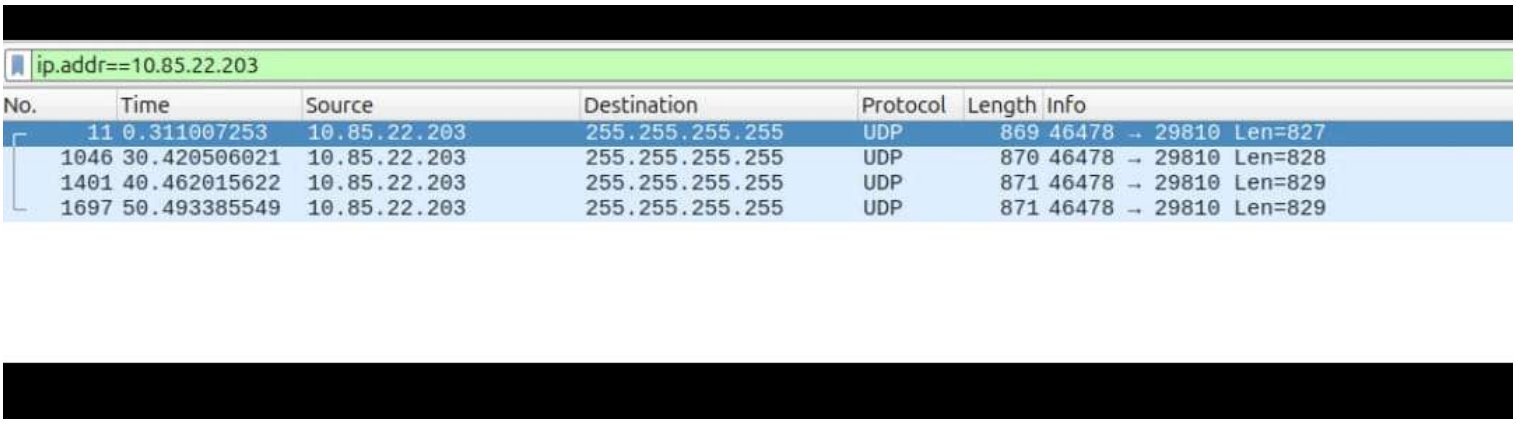
Q1 What are the different protocols you observe at the following layers at the Protocol Stack?

Application layer	Network layer	Transport layer
DNS	ARP	UDP
MDNS	$ICMPV_6$	TCP
DHCP		
HTTP		

Q.2 What is the total amount of data being received for the following two cases

A) when you access: http://reverssal.com/

B) when you access: https://www.cornell.edu


ip.addr==10.85.22.203
No. Time Source Destination Protocol Length Info
11 0.311007253 10.85.22.203 255.255.255.255 UDP 869 46478 → 29810 Len=827
1046 30.420506021 10.85.22.203 255.255.255.255 UDP 870 46478 → 29810 Len=828
1401 40.462015622 10.85.22.203 255.255.255.255 UDP 871 46478 → 29810 Len=829
1697 50.493385549 10.85.22.203 255.255.255.255 UDP 871 46478 → 29810 Len=829

ip.addr==10.85.19.164

No.	Time	Source	Destination	Protocol	Length	Info
111	2.356633322	10.85.19.164	255.255.255.255	UDP	871	47006 → 29810 Len=829
521	12.392992300	10.85.19.164	255.255.255.255	UDP	871	47006 → 29810 Len=829
909	22.425763910	10.85.19.164	255.255.255.255	UDP	872	47006 → 29810 Len=830
1297	32.460626205	10.85.19.164	255.255.255.255	UDP	872	47006 → 29810 Len=830
2829	72.597845281	10.85.19.164	255.255.255.255	UDP	872	47006 → 29810 Len=830
3180	82.634104137	10.85.19.164	255.255.255.255	UDP	870	47006 → 29810 Len=828

Q.3 Answer the following when you access the site http://neverssl.com/

A) How many HTTP GET request do you observe.

Ans. One HTTP GET request observed

GET requests:

Time	Source	Destination	Protocol	Length	Info
43893049906	10.85.16.239	34.223.124.45	HTTP	142	GET / HTTP/1.1

B) What version of HTTP the server is running

GET / HTTP/1.1


http
No. Time Source Destination Protocol Length Info
1499 43.939049906 10.85.16.239 34.223.124.45 HTTP 142 GET / HTTP/1.1

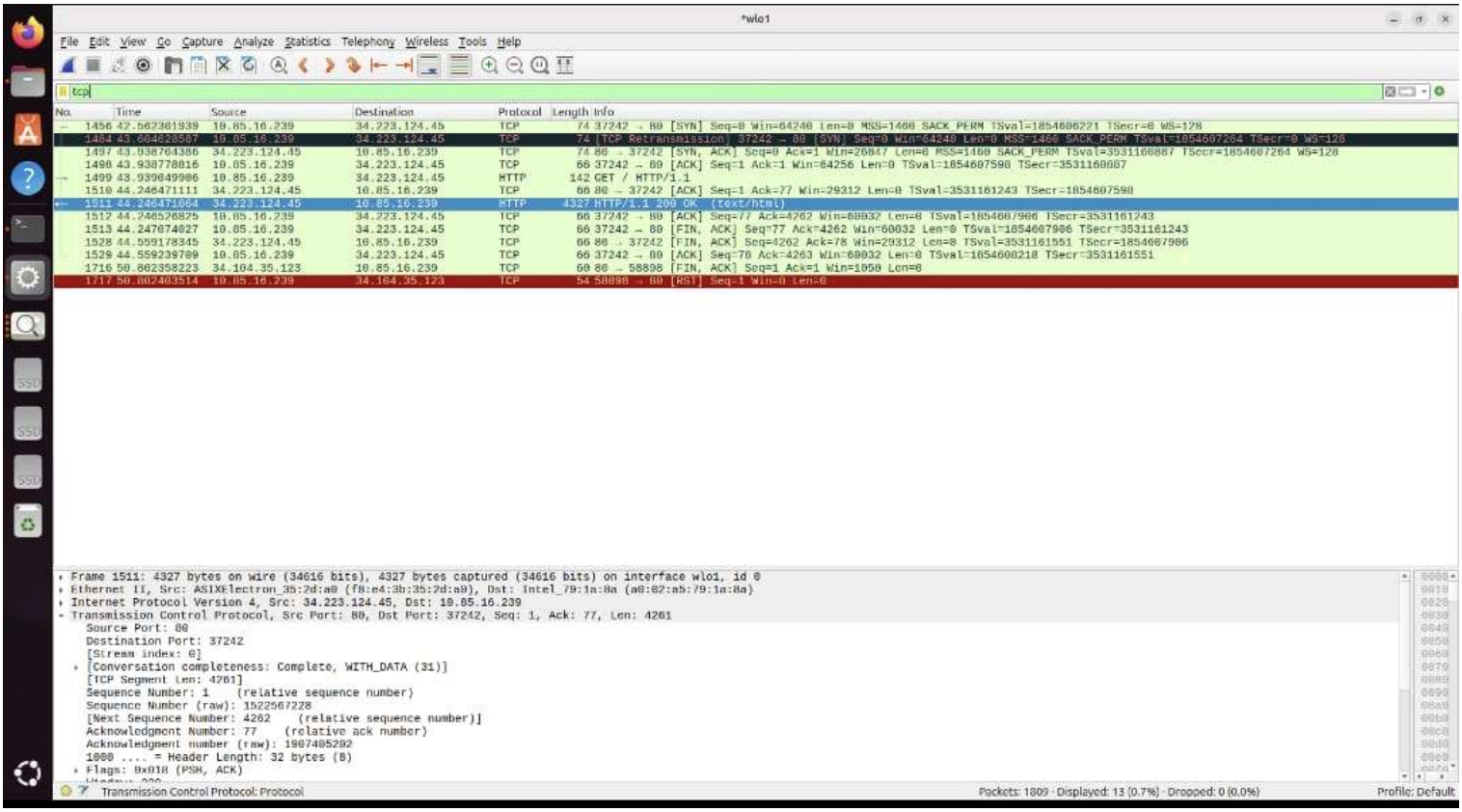
c) for each of the HTTP Get request find out;

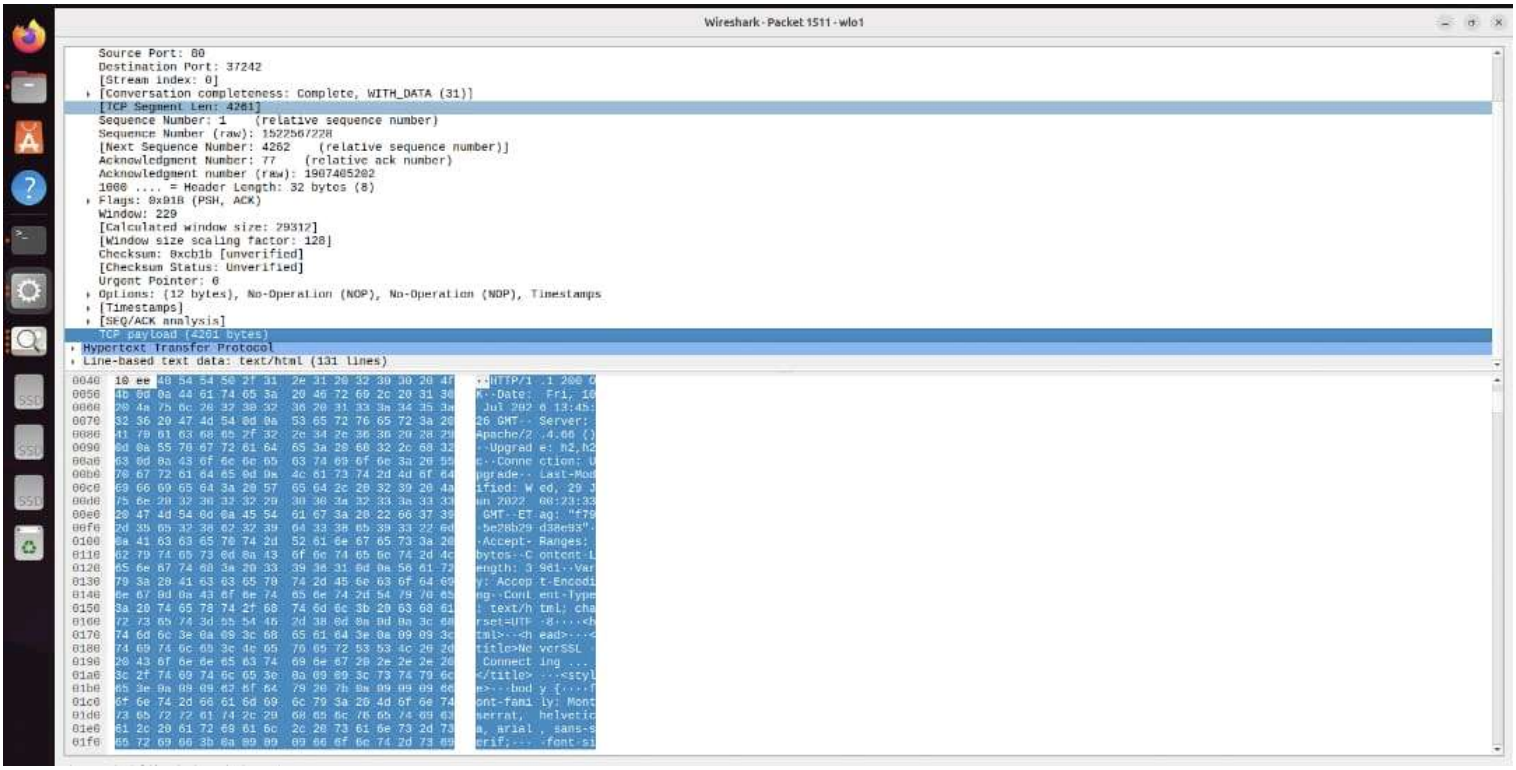
i) the total number of TCP segments being recieved

One TCP segment recieved

(ii) the total amount of data being recived int the corresponding HTTP response message

4261 byte


*wlo1
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tcp
No. Time Source Destination Protocol Length Info
1456 42.562361939 10.85.16.239 34.223.124.45 TCP 74 37242 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1854606221 TSecr=0 WS=128
1484 43.604628587 10.85.16.239 34.223.124.45 TCP 74 [TCP Retransmission] 37242 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1854607264 TSecr=0 WS=128
1497 43.938764386 34.223.124.45 10.85.16.239 TCP 74 80 → 37242 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1460 SACK_PERM TSval=3531160887 TSecr=1854607264 WS=128
1498 43.938778816 10.85.16.239 34.223.124.45 TCP 66 37242 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1854607598 TSecr=3531160887
1499 43.939649906 10.85.16.239 34.223.124.45 HTTP 142 GET / HTTP/1.1
1510 44.246471111 34.223.124.45 10.85.16.239 TCP 66 80 → 37242 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3531161243 TSecr=1854607598
1511 44.246471664 34.223.124.45 10.85.16.239 HTTP 4327 HTTP/1.1 200 OK (text/html)
1512 44.246526825 10.85.16.239 34.223.124.45 TCP 66 37242 → 80 [ACK] Seq=77 Ack=4262 Win=60032 Len=0 TSval=1854607906 TSecr=3531161243
1513 44.247074627 10.85.16.239 34.223.124.45 TCP 66 37242 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=60032 Len=0 TSval=1854607906 TSecr=3531161243
1528 44.559178345 34.223.124.45 10.85.16.239 TCP 66 80 → 37242 [FIN, ACK] Seq=4262 Ack=78 Win=29312 Len=0 TSval=3531161551 TSecr=1854607906
1529 44.559239709 10.85.16.239 34.223.124.45 TCP 66 37242 → 80 [ACK] Seq=78 Ack=4263 Win=60032 Len=0 TSval=1854608218 TSecr=3531161551
1716 50.802358223 34.104.35.123 10.85.16.239 TCP 60 80 → 58898 [FIN, ACK] Seq=1 Ack=1 Win=1050 Len=0
1717 50.802403514 10.85.16.239 34.104.35.123 TCP 54 58898 → 80 [RST] Seq=1 Win=0 Len=0
Frame 1511: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlo1, id 0
Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: Intel_79:1a:8a (a0:02:a5:79:1a:8a)
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.16.239
Transmission Control Protocol, Src Port: 80, Dst Port: 37242, Seq: 1, Ack: 77, Len: 4261
Source Port: 80
Destination Port: 37242
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 4261]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1522567228
[Next Sequence Number: 4262 (relative sequence number)]
Acknowledgment Number: 77 (relative ack number)
Acknowledgment number (raw): 1907485282
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Transmission Control Protocol: Protocol
Packets: 1809 · Displayed: 13 (0.7%) · Dropped: 0 (0.0%)
Profile: Default


Wireshark · Packet 1511 · wlo1
Source Port: 80
Destination Port: 37242
[Stream index: 0]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 4261]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1522567228
[Next Sequence Number: 4262 (relative sequence number)]
Acknowledgment Number: 77 (relative ack number)
Acknowledgment number (raw): 1907405202
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Window: 229
[Calculated window size: 29312]
[Window size scaling factor: 128]
Checksum: 0xcb1b [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
TCP payload (4261 bytes)
Hypertext Transfer Protocol
Line-based text data: text/html (131 lines)

D. Identify Tcp three way handshake used to establish the connection with the http://neverssl.com. Specify SYN, SYN-ACK packets and write the sequence number for each of the three accesses

SYN Packet:

Source IP: 10.85.17.18 (Client)

Destination IP: 34.223.124.45 (Server)

Protocol info: 57022 → 80 [SYN]

Sequence Number: 0

SYN-ACK Packet:

Source IP: 34.223.124.45 (Server)

Destination IP: 10.85.17.18 (Client)

Protocol info: 80 → 57022 [SYN-ACK]

Sequence Number: 0

(ACK: 1)

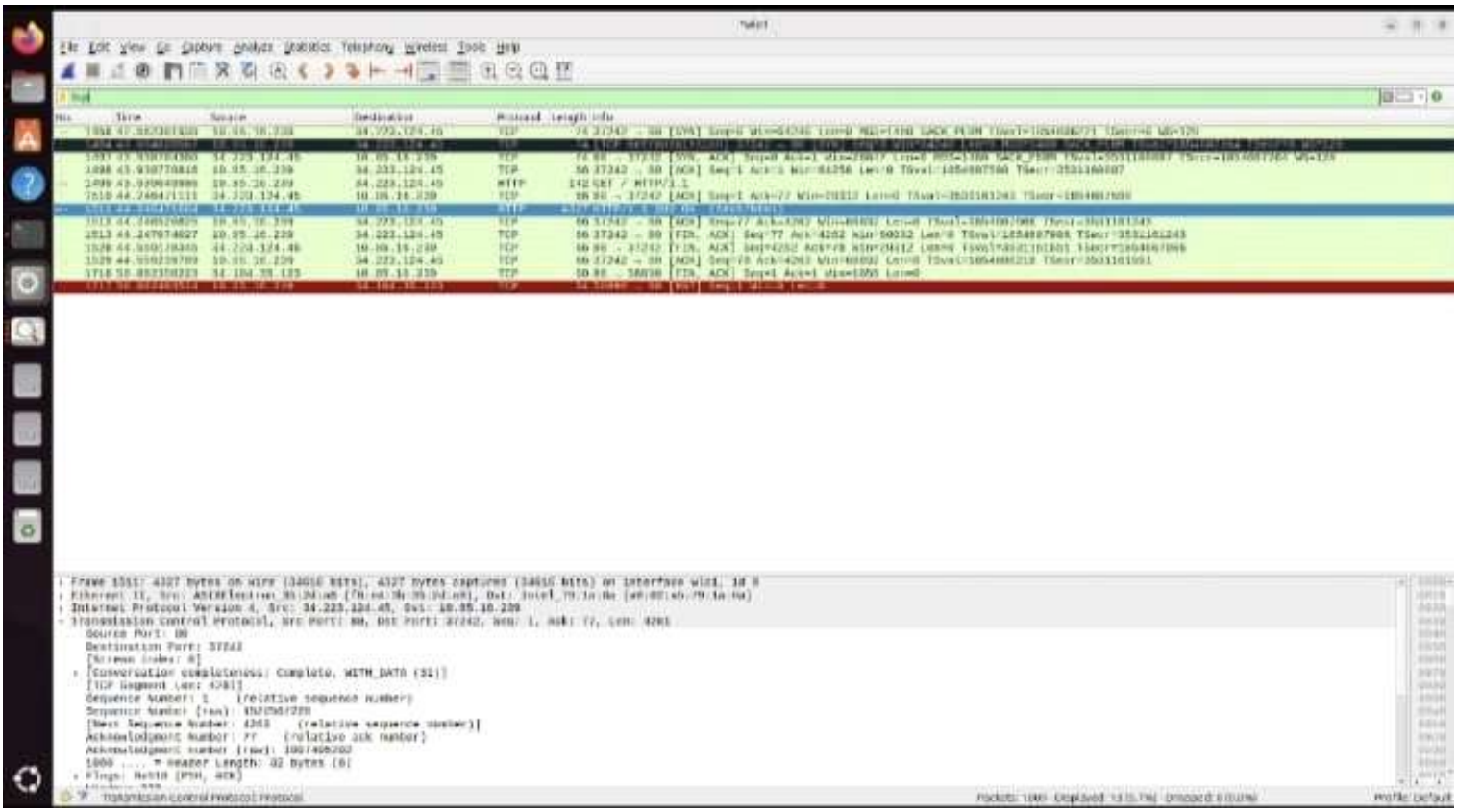
ACK Packet:

Source IP: 10.85.17.18 (Client)

Destination IP: 34.223.124.45 (Server)

Protocol info: 57022 → 80 [ACK]

Sequence Number: 1


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No. Time Source Destination Protocol Length Info
1508 47.557307880 10.95.10.239 34.223.124.45 TCP 74 37242 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1054988271 TSecr=0 WS=128
1509 47.557307880 10.95.10.239 34.223.124.45 TCP 74 [TCP Retransmission] 37242 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1054988271 TSecr=0 WS=128
1597 47.938704360 34.223.124.45 10.95.10.239 TCP 74 80 → 37242 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1360 SACK_PERM TSval=3531181087 TSecr=1054987761 WS=128
1598 47.938770816 10.95.10.239 34.223.124.45 TCP 66 37242 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=1054987760 TSecr=3531180987
1599 47.939040980 10.95.10.239 34.223.124.45 HTTP 142 GET / HTTP/1.1
1610 48.246471111 34.223.124.45 10.95.10.239 TCP 66 80 → 37242 [ACK] Seq=1 Ack=77 Win=28933 Len=0 TSval=3531181243 TSecr=1054988069
1611 48.246520825 10.95.10.239 34.223.124.45 TCP 66 37242 → 80 [ACK] Seq=77 Ack=4262 Win=60032 Len=0 TSval=1054988069 TSecr=3531181243
1613 48.247074027 10.95.10.239 34.223.124.45 TCP 66 37242 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=60032 Len=0 TSval=1054988069 TSecr=3531181243
1628 48.550139445 34.223.124.45 10.95.10.239 TCP 66 80 → 37242 [FIN, ACK] Seq=4262 Ack=78 Win=28412 Len=0 TSval=3531181551 TSecr=1054988069
1629 48.550239709 10.95.10.239 34.223.124.45 TCP 66 37242 → 80 [ACK] Seq=78 Ack=4263 Win=60032 Len=0 TSval=1054988218 TSecr=3531181551
1716 50.853350223 52.184.35.125 10.95.10.239 TCP 60 80 → 58038 [FIN, ACK] Seq=1 Ack=1 Win=1055 Len=0
1717 50.853405513 10.95.10.239 52.184.35.125 TCP 54 58038 → 80 [RST] Seq=1 Win=0 Len=0
Frame 1511: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlo1, id 0
Ethernet II, Src: AzureWaveTec_35:2d:a6 (70:66:55:35:2d:a6), Dst: Intel_79:1a:0a (a0:60:b6:79:1a:0a)
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.95.10.239
Transmission Control Protocol, Src Port: 80, Dst Port: 37242, Seq: 1, Ack: 77, Len: 4261
Source Port: 80
Destination Port: 37242
[Stream index: 6]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 4261]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 892565720
[Next Sequence Number: 4263 (relative sequence number)]
Acknowledgment Number: 77 (relative ack number)
Acknowledgment number (raw): 3531405293
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Transmission Control Protocol: Protocol
Packets: 1980 · Displayed: 13 (0.7%) · Dropped: 0 (0.0%)
Profile: Default